

EM 651: Photovoltaic materials and fabrication

Course Code	EM 651
Course Name	Photovoltaic materials and fabrication
L-T-P-C	3-0-0-3
Prerequisites	Device electronics for integrated circuits (EE 31 I)/ Introductory course on semiconductor devices/Instructor's consent
Intended for	3rd year and 4th year UG, PG
Distribution	Elective for ME
One-time approval	14th BoA (Sep 2016); 11th Senate (Oct 2016)

Course Contents

- **Semiconductor Electronics** [5 Lectures]
 - Physics of Semiconductor Materials
 - Band Model of Solids
 - Concept of equilibrium and non-equilibrium in semiconductor device
 - Current conduction mechanisms in semiconductors
 - Fundamentals of p-n junction
 - Operation of p-n junction under forward and reverse bias
- **Basics of Photovoltaics** [7 Lectures]
 - History of solar cells
 - Principle of detailed balance
 - Principles of carrier separation and generation-recombination
 - Operation of solar cells
 - Design of solar cells
- **Advanced concepts in solar cells** [7 lectures]
 - Theory of multi-junctions solar cells with an emphasis on tunnel junctions between them
 - Physics of heterojunctions and other junctions such as Schottky diodes and ohmic contacts
 - Induced junctions, MIS solar cell and carrier selective contact approaches
- **Solar resources and PV performance** [4 Lectures]
 - Solar radiation
 - Calculation of direct and diffuse radiation
 - Solar modules and arrays
 - Performance evaluation of PV modules
- **Reliability of photovoltaic modules** [6 Lectures]
 - Familiarization with IEC certification testing
 - Microclimate of PV module
 - Influence of microclimate on reliability of PV module
- **Components of photovoltaic systems** [4 Lectures]
 - Different PV System Topology
 - Introduction to different power electronic components
 - Grid integration of photovoltaics
- **PV Technology** [7 Lectures]
 - Silicon solar cell fabrication
 - III-V technology
 - Thin film technology
 - Other emerging technologies

Text Books

1. Jenny Nelson, **Physics of solar cells**, Imperial College Press, 2003
2. Roger Messenger, Jerry Ventre, **Photovoltaic systems engineering**, CRC press